

## RESEARCH INTERESTS

I study **foundation models**: what they learn, how to interpret and steer them, and how to extend them beyond text.

- **Algorithmic perspectives on LLMs**: whether Transformers learn algorithms from data [C4, C14], how pretrained LLMs do arithmetic [C3, C10, C18], and what sets Transformers apart from other architectures [C7].
- **Interpretability and alignment**: decision making under uncertainty [C6, C15], steering vectors [C12, C13], and mechanistic interpretability with SAEs and transcoders [C16, T1].
- **Foundation models beyond text**: multimodal reward models [C8, C5], modality sensitivity [C2], data for visual reasoning [C11], and tabular foundation models [T2, P4].

## EDUCATION

**University of Southern California** 2022 – Present

*Doctor of Philosophy (Ph.D.) in Computer Science*

*Advisors: Vatsal Sharan & Robin Jia*

**University of Chicago** 2020 – 2022

*Master of Science (S.M.) in Statistics*

**University of Chicago** 2016 – 2020

*Bachelor of Science (S.B.) in Mathematics with Honors*

*Bachelor of Science (S.B.) in Computer Science with Honors*

*Bachelor of Arts (A.B.) in Statistics*

## EXPERIENCE

**Google Research** May 2026 – Present

*Student Researcher* • Hosts: Weihao Kong, Abhimanyu Das

**Simons Institute at UC Berkeley** Fall 2024

*Visiting Graduate Student* • Program: Modern Paradigms in Generalization

**Meta Llama Team** May – Dec., 2024

*AI Research Scientist Intern* • Host: Lawrence Chen; Peers: Tong Xiao, Rui Wang, and Pengchuan Zhang

**Google** May – Aug., 2023

*Software Engineering Intern* • Team: Bard (now Gemini) Multilingual

## PUBLICATIONS

featured \* equal contribution † mentored student

[C19] **Value-Aware Stochastic KV Cache Eviction for Reasoning Models**

Ting-Yun Chang\*, Harvey Yiyun Fu\*, Deqing Fu\*, Chenghao Yang, Jesse Thomason, Robin Jia

*Neural Information Processing Systems (NeurIPS), 2026*

[C18] **Convergent Evolution: How Different Language Models Learn Similar Number Representations**

Deqing Fu, Tianyi Zhou, Mikhail Belkin, Vatsal Sharan, Robin Jia

*Conference on Language Modeling (COLM), 2026*

[C17] **Token-Level Off-Policy Learning for Faithful Generation Under Distribution Shift**

Cynthia Huang\*†, Gustavo Lucas Carvalho\*†, Deqing Fu, Robin Jia

*Conference on Language Modeling (COLM), 2026*

[C16] **Resa: Transparent Reasoning Models via SAEs**

Shangshang Wang, Julian Asilis, Ömer Faruk Akgül, Enes Burak Bilgin, Ollie Liu, Deqing Fu, Willie Neiswanger

*Conference on Language Modeling (COLM), 2026*

- [C15] **Are LLM Decisions Faithful to Verbal Confidence?**  
Jiawei Wang, Yanfei Zhou, Siddartha Devic, Deqing Fu  
*Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026
- [C14] **Transformers Provably Learn Algorithmic Solutions for Graph Connectivity, But Only with the Right Data**  
Qilin Ye<sup>\*†</sup>, Deqing Fu<sup>\*</sup>, Robin Jia, Vatsal Sharan  
*International Conference on Machine Learning (ICML)*, 2026
- [C13] **EPSVec: Efficient and Private Synthetic Text Generation via Dataset Vectors**  
Amin Banayeeanzade, Qingchuan Yang, Deqing Fu, Spencer Hong, Erin Babinsky, Alf Samuel, Anoop Kumar, Robin Jia, Sai Praneeth Karimireddy  
*International Conference on Machine Learning (ICML)*, 2026
- [C12] **Textual Steering Vectors Can Improve Visual Understanding in Multimodal Large Language Models**  
Woody Haosheng Gan<sup>\*†</sup>, Deqing Fu<sup>\*</sup>, Julian Asilis<sup>\*</sup>, Ollie Liu<sup>\*</sup>, Dani Yogatama, Vatsal Sharan, Robin Jia, Willie Neiswanger  
*Association for Computational Linguistics (ACL)*, 2026
- [C11] **Zebra-CoT: A Dataset for Interleaved Vision Language Reasoning**  
Ang Li<sup>\*</sup>, Charles L. Wang<sup>\*</sup>, Deqing Fu<sup>\*</sup>, Kaiyu Yue<sup>\*</sup>, Zikui Cai<sup>\*</sup>, Wang Bill Zhu<sup>\*</sup>, Ollie Liu<sup>\*</sup>, Peng Guo<sup>\*</sup>, Willie Neiswanger, Furong Huang, Tom Goldstein, Micah Goldblum  
*International Conference on Learning Representations (ICLR)*, 2026
- [C10] **FoNE: Precise Single-Token Number Embeddings via Fourier Features**  
Tianyi Zhou, Deqing Fu, Mahdi Soltanolkotabi, Robin Jia, Vatsal Sharan  
*International Conference on Learning Representations (ICLR)*, 2026
- [C9] **VisualLens: Personalization through Task-Agnostic Visual History**  
Wang Bill Zhu, Deqing Fu, Kai Sun, Yi Lu, Zhaojiang Lin, Seungwhan Moon, Kanika Narang, Mustafa Canim, Yue Liu, Anuj Kumar, Xin Luna Dong  
*Neural Information Processing Systems (NeurIPS)*, 2025
- [C8] **TLDR: Token-Level Detective Reward Model for Large Vision Language Models**  
Deqing Fu, Tong Xiao, Rui Wang, Wang Zhu, Pengchuan Zhang, Guan Pang, Robin Jia, Lawrence Chen  
*International Conference on Learning Representations (ICLR)*, 2025
- [C7] **Transformers Learn Low Sensitivity Functions: Investigations and Implications**  
Bhavya Vasudeva<sup>\*</sup>, Deqing Fu<sup>\*</sup>, Tianyi Zhou, Elliot Kau<sup>†</sup>, You-Qi Huang<sup>†</sup>, Vatsal Sharan  
*International Conference on Learning Representations (ICLR)*, 2025
- [C6] **DeLLMa: Decision Making Under Uncertainty with Large Language Models**  
Ollie Liu<sup>\*</sup>, Deqing Fu<sup>\*</sup>, Dani Yogatama, Willie Neiswanger  
*International Conference on Learning Representations (ICLR)*, 2025  
*ICLR 2025 Spotlight (Top 5.1%)*
- [C5] **DreamSync: Aligning Text-to-Image Generation with Image Understanding Feedback**  
Jiao Sun<sup>\*</sup>, Deqing Fu<sup>\*</sup>, Yushi Hu<sup>\*</sup>, Su Wang, Royi Rassin, Da-Cheng Juan, Dana Alon, Charles Herrmann, Sjoerd van Steenkiste, Ranjay Krishna, Cyrus Rashtchian  
*Nations of the Americas Chapter of the ACL (NAACL)*, 2025
- [C4] **Transformers Learn to Achieve Second-Order Convergence Rates for In-Context Linear Regression**  
Deqing Fu, Tian-Qi Chen, Robin Jia, Vatsal Sharan  
*Neural Information Processing Systems (NeurIPS)*, 2024  
*SoCalNLP Symposium 2023 Best Paper Award*
- [C3] **Pre-trained Large Language Models Use Fourier Features to Compute Addition**  
Tianyi Zhou, Deqing Fu, Vatsal Sharan, Robin Jia  
*Neural Information Processing Systems (NeurIPS)*, 2024
- [C2] **IsoBench: Benchmarking Foundation Models on Isomorphic Representations**  
Deqing Fu<sup>\*</sup>, Ruohao Guo<sup>\*</sup>, Ghazal Khalighinejad<sup>\*</sup>, Ollie Liu<sup>\*</sup>, Bhuwan Dhingra, Dani Yogatama, Robin Jia, Willie Neiswanger  
*Conference on Language Modeling (COLM)*, 2024

[C1] **SCENE: Self-Labelled Counterfactuals for Extrapolating to Negative Examples**

Deqing Fu, Ameya Godbole, Robin Jia  
*Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023

**PRE-PRINTS, TECH REPORTS, AND BLOG POSTS**      **featured**    \* equal contribution    † mentored student

- [P4] **TabFM-Auto: Self-Evolving Pipelines for Tabular Foundation Models**  
Deqing Fu, Huangyuan Su, Rajat Sen, Taman Narayan, Sujay Sanghavi, Abhimanyu Das, Weihao Kong
- [P3] **Rewards Beyond Test Passing: Towards Generalizable Fidelity in LLM Debugging**  
Yunkai Zhan, Miaosen Chai, Shangshang Wang, Jike Zhong, Yuqing Yang, Deqing Fu, Serina Chang, Wang Bill Zhu
- [P2] **Learning to GUIDE the Group: Self-Learning LLM Facilitation Across Multi-Party Tasks**  
Ziyi Liu, Tzu-Sheng Kuo, Jiale Liu, Deqing Fu, Nithum Thain, Crystal Qian
- [P1] **Unsupervised Tool Evolution for Coding Agents in Data-Lake Analytics**  
Yuqing Yang, Yurong Liu, Deqing Fu, Manav Garg, Sravan Babu Bodapati, Fatma Ozcan, Alon Halevy, Jiani Zhang
- [T2] **TabFM: A Zero-shot Foundation Model for Tabular Data** *Technical Report*  
Weihao Kong, Erez Louidor Ilan, Shuxin Nie, Taman Narayan, Rajat Sen, Yichen Zhou, Deqing Fu, Samet Oymak, Abhimanyu Das
- [T1] **Qwen3 Cross-layer Transcoders** *Technical Report*  
Jakob Hansen, Deqing Fu, Shangshang Wang, John Carlsson
- [B1] **Seeing Is Not Reasoning: How VLMs and Their Benchmarks Lean on Text** *Blog Post*  
Harvey Yiyun Fu\*, Chenghao Yang\*, Ting-Yun Chang, Zhaowei Wang, Zhaofeng Wu, Jiawei Zhou, Deqing Fu

**AWARDS**

|                                                                            |                     |
|----------------------------------------------------------------------------|---------------------|
| <b>ACL Outstanding Reviewer Award</b>                                      | ACL, 2026           |
| <b>NeurIPS Top Reviewer</b>                                                | NeurIPS, 2025       |
| <b>Viterbi Best Research Assistant Award</b> ( <i>one per department</i> ) | USC, 2025           |
| <b>USC Capital One Center (CREDIF) PhD Fellow</b>                          | USC, 2025           |
| <b>SoCalNLP Symposium 2023 Best Paper Award</b>                            | SoCalNLP, 2023      |
| <b>Provost’s Fellowship</b>                                                | USC, 2022           |
| <b>Susanne H. Rudolph Scholarship</b>                                      | UChicago, 2020–2021 |
| <b>Liew Family College Research Fellows Fund</b>                           | UChicago, 2020      |
| <b>Dean’s List</b>                                                         | UChicago, 2016–2020 |

**INVITED TALKS**

|                                                                                                     |                          |
|-----------------------------------------------------------------------------------------------------|--------------------------|
| <b>Emergent Algorithms in Foundation Models: Theory, Discovery, and Design</b>                      |                          |
| Columbia NLP Seminar                                                                                | Nov 5, 2026 (scheduled)  |
| Cornell Tech ML Seminar                                                                             | Nov 3, 2026 (scheduled)  |
| Harvard ML Foundations Seminar                                                                      | Oct 28, 2026 (scheduled) |
| ISI Natural Language Seminar                                                                        | Sep 22, 2026             |
| USC NLP Lunch                                                                                       | Sep 10, 2026             |
| <b>Theoretical Foundations of LLMs: Emergent Algorithms and Model Design</b>                        |                          |
| Google Research Seminar                                                                             | Aug 6, 2026              |
| <b>Algorithmic Perspectives on Understanding Transformers</b>                                       |                          |
| FlaNN Talk Seminar                                                                                  | Aug 25, 2025             |
| UChicago/TTIC NLP Seminar                                                                           | May 08, 2025             |
| Duke NLP Seminar                                                                                    | Feb 12, 2025             |
| <b>Closing the Modality Gap: Benchmarking and Improving Visual Understanding in Multimodal LLMs</b> |                          |
| Stanford NLP Seminar                                                                                | May 22, 2025             |

## Simplicity Bias of Transformers to Learn Low Sensitivity Functions

USC NLP Lunch

Feb 22, 2024

## Aligning Text-to-Image Generation with Image Understanding Feedback

USC NLP Lunch

Oct 19, 2023

## Transformers Learn to Achieve Second-Order Convergence for In-Context Linear Regression

USC NLP Lunch

Mar 29, 2023

## SERVICES

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**Reviewer** for conferences in

- **Machine Learning:** ICLR (2025–2027), NeurIPS (2024–2026), ICML (2025)
- **Natural Language Processing:** COLM (2024–2026), EMNLP (2023), ACL Rolling Review (2024–2026)
- **Computer Vision:** CVPR (2023–2025), ECCV 2022, ICCV 2025

**Organizer** of USC ML/NLP Joint Reading Group

Jan 2025 – May 2025

**Organizer** of USC AI Safety Club

Jan 2024 – Jan 2025

## STUDENTS SUPERVISED

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### Undergraduate Students

Zitong Cynthia Huang (USC UG)

Sep 2025 – May 2026

*Now: Ph.D. Student in Electronic and Computer Engineering at University of Southern California*

Qilin Ye (USC UG)

Sep 2023 – Feb 2026

*Now: Ph.D. Student in Computer Science at Stanford University*

Woody Haosheng Gan (USC UG)

Sep 2023 – May 2026

*Now: Ph.D. Student in Computer Science at Georgia Institute of Technology*

You-Qi Huang (USC UG)

Sep 2023 – Mar 2024

### Master's Students

Sophia Mirrashidi (USC MS)

Jan 2026 – Present

Gustavo Lucas Carvalho (USC MS)

Jan 2025 – May 2026

Elliot Kau (USC MS)

Sep 2023 – Mar 2024

*Now: Software Engineer at Google*

Tian-Qi Chen (USC MS)

Mar 2023 – Sep 2023

## TEACHING EXPERIENCE

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**Teaching Assistant**, CSCI 467: Introduction to Machine Learning

Spring 2025

University of Southern California, Los Angeles, CA

**Teaching Assistant**, CSCI 566: Deep Learning and its Applications

Spring 2023

University of Southern California, Los Angeles, CA

**Teaching Assistant**, BUSN 41301: Statistical Insight

Fall 2021

Booth School of Business, University of Chicago, Chicago, IL

**Course Assistant**, CMSC 31230: Fundamentals of Deep Learning

Fall 2021

University of Chicago, Chicago, IL

**Course Assistant**, MATH 20300: Analysis in  $\mathbb{R}^n$  – I

Spring 2018

University of Chicago, Chicago, IL

## SKILLS

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**Languages:** C/C++, Python, L<sup>A</sup>T<sub>E</sub>X, MATLAB, R, Java, CUDA

**Frameworks:** PyTorch, TensorFlow, Keras, JAX, HuggingFace